

Development Of A Hybrid Deep Learning Framework For Predicting Epileptic Seizures Using EEG Signal Analysis.

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ABSTRACT Epilepsy impacts the lives of over 65 million people worldwide, with many cases resistant to medication or surgical intervention. Early seizure prediction can offer a critical opportunity for therapeutic response and improved quality of life. This paper proposes a hybrid deep learning framework to predict seizures by analyzing EEG signal patterns. The proposed method consists of pre-processing the scalp EEG signals, extracting features presenting the extracted features using CNN and classifying the features with SVM resulting in an accuracy of 98.14%. To enhance robustness, an ensemble architecture combining CNN, LSTM, SqueezeNet, and ANN is introduced. This is supported by advanced preprocessing techniques like bandpass filtering, ICA, and Z-score normalization. The system supports multi-class detection and is designed for integration with telehealth platforms, allowing real-time deployment in clinical settings. Experimental results confirm the model's effectiveness in accurate and timely seizure prediction, highlighting its potential for use in intelligent epilepsy management systems.

INDEX TERMS Epilepsy prediction, Interictal State, Preictal State, Independent Component Analysis, CNN, ANN, LSTM, SqueezeNet, Hybrid Chimpanzee Enhanced Fox Optimization (HCEFO) algorithm.

I. INTRODUCTION

Epilepsy is Brain disorder marked by recurrent seizures resulting from excessive or disorganized electrical pulses in brain. The disorder affects over 1% of global population, and many of them still have uncontrolled seizures despite advances in available pharmacological and surgical treatments [21]. In fact, more than 30% of patients have drug resistant epilepsy, in which their seizures cannot be prevented by conventional therapies [22]. Given this, it is important to develop predictive models that can warn of an impending seizure and provide an opportunity to take preventive action.

Seizure prediction systems work with brain signals available through electroencephalogram (EEG), which can be acquired non-invasively with scalp electrodes (scalp EEG) or invasively with intracranial electrodes (iEEG), on the basis of the clinical situation. EEG signals capture different states of seizure activity. These include interictal (or normal activity of brain), preictal (period prior to seizure), ictal (amid seizure), and postictal (post seizure). The preictal state is typically defined as the 30-90 minutes prior to the seizure and represents a critical time period for prediction models. If detection can occur during the preictal state, intervention through therapy (e.g., medication) or neurostimulation can be facilitated [23].

Digital EEG recordings are obtained via headsets or specialized EEG acquisition systems to register the brain's electrical activity with sampling frequencies also known as sample rate, between 200 Hz and 5000 Hz. As part of routine clinical practice neurologists annotate digital EEG recordings using clinical software to indicate the seizure event and transitions between physiological states. EEG signals show a distinct morphology in interictal, preictal and ictal states which can be effectively distinguished using pattern recognition methods.

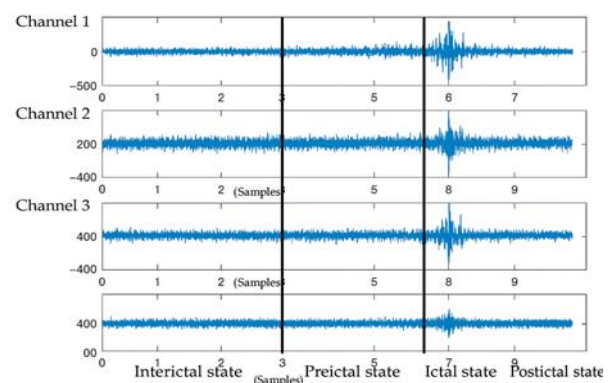


FIGURE 1. Overview of seizure states across three EEG channels recorded for 1 hour, depicting interictal, preictal, ictal, and postictal phases.

Recent developments in machine-learning and deep-learning have greatly enhanced prediction accuracy along with reliability of seizure prediction models. In general, the modeling pipeline incorporates pre-processing, including to remove noise and artefacts, feature extraction to quantify EEG signals static, spectral, or temporal properties, and classification to detect preictal patterns. Pre-processing is commonly completed using bandpass and notch filters, common spatial pattern (CSP) filtering, or Empirical Mode Decomposition (EMD) processes that remove irrelevant frequency components. Feature extraction can be completed by quantifying signal entropy, power spectral densities, Lyapunov exponents, and the other temporal descriptors. Traditional deep learning models such as CNNs and LSTM networks allow automation of feature learning and classification. Further, they support end-to-end seizure prediction paradigms.

In the paper, we provide deep learning solutions to seizure prediction on the basis of scalp EEG recordings. The subsequent sections are detailed as follows: Section II presents relevant literature allied to EEG-based seizure prediction, Section III offers methodology, Section IV presents result discussion, and Section V yields the conclusion and future work, including a discussion and possible future work.

II. LITERATURE REVIEW

Epileptic seizure prediction systems typically follow a structured pipeline that begins with preprocessing of EEG signals, followed by feature extraction, and finally classification. Numerous studies have explored this approach, particularly using scalp EEG, where electrodes are placed externally to monitor brain activity. In recent years, a growing number of researchers [1]–[20] have designed and implemented systems for predicting seizures using machine-learning and deep-learning techniques rely on scalp EEG signal patterns. These approaches generally share a common framework that includes cleaning and preparing the raw signals, identifying distinctive patterns or features, and classifying the data into preictal and interictal states.

A. PRE-PROCESSING

The quality of EEG signals captured from patients is often compromised by various types of noise introduced during acquisition. These unwanted disturbances—such as power line interference (typically between 50 and 60 Hz), electrode-related artifacts, and physiological noises from eye movement or heartbeat—can significantly degrade the signal-to-noise ratio (SNR). This degradation can negatively affect the accuracy of seizure prediction models by making it difficult to distinguish between seizure-prone and normal states.

To improve signal clarity, various preprocessing strategies have been proposed in the literature. Traditional methods like bandpass and notch filtering are commonly used to eliminate known frequency-based noise.

Additionally, low-pass and high-pass filters can be used to suppress low-frequency drifts and high-frequency noise, respectively. More complex methods are available in frequency-domain, with the FFT and STFT designed to analyze the spectral components of non-stationary EEG data. Other sophisticated methods like Wavelet Transform and Empirical Mode Decomposition (EMD) allow for multi-resolution analysis of EEG signals, effectively separating noise from the signal across various frequency bands.

In studies such as [9] and [13], these advanced preprocessing methods were crucial in enhancing the quality of EEG input for subsequent feature extraction and classification. Surrogate channel approaches and spatial filtering techniques, including Common Spatial Patterns (CSP), were also employed in [11] and [15] to suppress artifacts and improve signal clarity. These techniques collectively ensure that the EEG signals used for analysis retain their meaningful components, which is vital for reliable prediction.

B. FEATURE EXTRACTION

After the EEG data is cleaned and refined, the next crucial step is to extract meaningful features that can help distinguish between different brain states. Feature extraction strategies can broadly be categorized into two types: handcrafted and automated.

Handcrafted features are derived from expert knowledge and statistical analysis of the EEG signal. These include temporal characteristics like mean, variance, skewness, and kurtosis, as well as more complex indicators such as entropy measures, Lyapunov exponents, and Principal Component Analysis (PCA). In the frequency domain, features like power spectral density and spectral moments are commonly used. Studies such as [1], [6], and [10] have successfully applied these techniques to capture the signal's dynamics and enhance model interpretability.

Automated feature extraction, often associated with deep learning, relies on neural networks to learn important features directly from raw or minimally processed data. CNNs are especially effective at identifying spatial rhythm in EEG signals, while models like LSTM and BiLSTM can capture temporal dependencies. Hybrid models that combine CNNs with recurrent architectures have been particularly effective in recent research [3], [9], [12].

Attention approaches, in addition to feature selection strategies described in [4] and [13], help prune the feature space by allowing us to focus on the most salient aspect of the EEG signal. These methods enhance model performance and interpretability, especially in multi-patient datasets or real-time applications.

C. CLASSIFICATION

After features are extracted, either manually or through automated methods, the final step is to classify the EEG data into interictal (non-seizure) or preictal (pre-seizure) states. This classification forms the basis of seizure prediction systems and has been tackled using both classical machine learning and modern deep-learning practices. Traditional classifiers like SVM, Random Forest (RF), k-Nearest Neighbors (KNN), Naive Bayes, and Gaussian Mixture Models (GMM) have shown consistent performance across various datasets [1], [5], [6], [16]. These models often require well-defined features and benefit from dimensionality reduction techniques like PCA, as demonstrated in [12] and [17].

Deep learning classifiers have gained prominence, because they can detect intricate nonlinear patterns in EEG. Architectures such as CNNs, RNNs, and LSTMs have been widely adopted for their high accuracy and robustness. In [8], [10], and [14], deep networks were used not only to classify seizure states but also to identify multiple seizure types with improved specificity and sensitivity. Hybrid and ensemble approaches have also emerged as powerful solutions. For instance, [8] combined deep learning with traditional ensemble classifiers like XGBoost and SVM, producing results above 98% accuracy; or in [11], the trio CNN, MLP and SVM resulted in an increase the robustness of the prediction system, particularly in terms of cross-validation.

Recent works [20] illustrate how hyperparameter tuning can affect model performance. For instance, SVM achieved high accuracy of almost 98% with hyperparameter tuning via randomized search. These models demonstrate that thorough model selection and tuning can improve seizure prediction accuracy and reliability substantially.

In summary, the literature demonstrates significant advancements in EEG-based seizure prediction systems through the effective use of preprocessing, feature extraction, and classification techniques. These combined efforts help develop smart, reliable, and personalized healthcare tools for managing epilepsy.

III. PROPOSED METHOD

We present a seizure predicting model that predicts the preictal state onset minutes before the seizure finally happens. The electroencephalography (EEG) signals used in the study came from publicly available CHB-MIT dataset and recorded 24 subjects, with 23 electrodes and a 256 Hz sampling rate, demonstrated in Figure 2. First the EEG files were converted to Matlab (`.mat`) format using the `edfread` function, and all data were loaded into `Matlab` to retrieve the signals. After loading the signals into Matlab, a Butterworth bandpass filter was applied to each signal to eliminate baseline drift and power line noise from the signal, ensuring quality signal data before proceeding.

To further enhance signal-to-noise ratio and provide time-frequency depiction, STFT is applied using non-overlapping 30-second windows, converting the time-domain signals into spectrograms of 256×256 pixels. These spectrograms are used as input to a custom three-layer Convolutional Neural Network (CNN), which uses 3×3 kernels, ReLU activations, and pooling layers to extract meaningful spatial features. Finally, instead of using fully connected layers, a linear SVM is used for classifying the EEG segments into interictal and preictal states. To enhance performance.

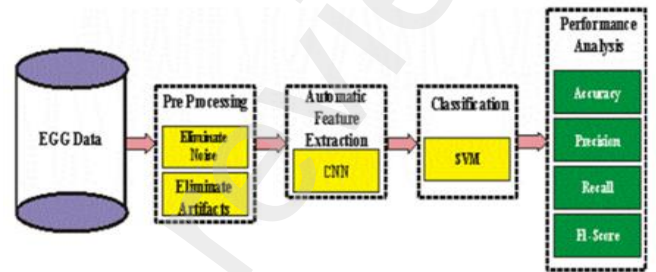


FIGURE 2. Flow Diagram for CNN-SVM Model

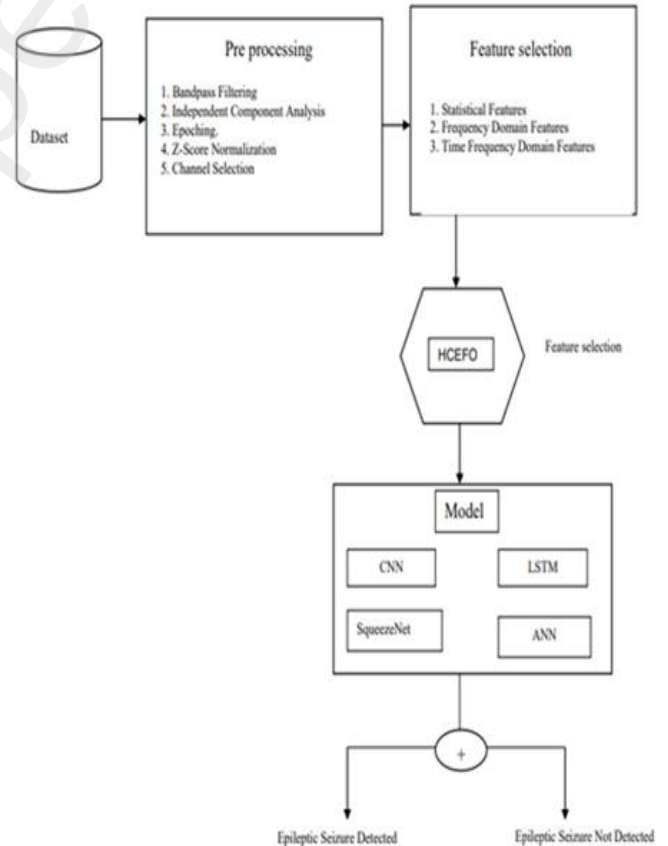


FIGURE 3. Architecture to analyse system using various algorithms.

Additional preprocessing steps such as Independent Component Analysis (ICA), Z-score normalization, and Common Spatial Pattern (CSP)-based channel selection are integrated. Also, the model use different neural architectures to enhance performance and effectively capture temporal dependencies in EEG data. The various architectures included LSTM networks to model temporal dependencies in EEG data, Artificial Neural Networks (ANN) to model complex nonlinear relationships, and SqueezeNet to reduce model size without sacrificing accuracy as shown in Figure. 3. The hybrid model used the strengths of both supervised deep and unsupervised classical learner classifier/model to predict epileptic seizures with accuracy, efficiency and real-time capabilities. The upcoming subsections provide brief overview of CNN–SVM Hybrid Model, ANN, LSTM, and SqueezeNet.

A. CNN–SVM Hybrid Model

CNNs are extensively used for feature extraction and classification of time series data and images. They comprise multiple layers that perform convolution followed by pooling, culminating in fully connected layers or external classifiers for final decision-making. In this hybrid approach, we combine feature extraction capability of CNN with classification capability of Support Vector Machines (SVM) for enhanced performance on epileptic EEG data.

$$W'(t+1) = W'(t) - \eta \cdot \nabla_{W'} C(W'(t))$$

$$B'(t+1) = B'(t) - \eta \cdot \nabla_{B'} C(B'(t))$$

Where:

- W', B' : Weight and biases in l^{th} layer
- η : Learning rate
- ∇C : Gradient of cost function with respect to the weights or biases
- t : Training iteration

Instead of using dense neural layers for classification, the hybrid model employs SVM to distinguish between preictal and interictal classes. SVM is supervised learning algorithm that identifies optimal separating hyperplane between two classes by maximizing margin between support vectors.

The decision function for linear SVM is defined as:

$$f(x) = w^T x + b$$

Where:

- x : Input feature vector (from CNN)
- w : Weight vector
- b : Bias term

B. ANN

ANN is a flexible and adaptable model derived from the human brain. ANN consists of several layers of interconnected processing units known as neurons. Each neuron computes weighted aggregate of its inputs and applies an activation function. ANN is ideal for capturing non-linear relationships in input data, making it particularly effective for epilepsy seizure prediction.

The basic operation of an ANN neuron can be mathematically represented as:

$$y = f\left(\sum_{i=1}^n w_i x_i + b\right)$$

where x_i denotes input features, w_i corresponding weights, b is bias term and f is non-linear activation function such as sigmoid, tanh, or rectified linear unit (ReLU).

In this proposed method, ANN is used for classification after extracting key features. It plays a vital role in learning complex seizure-related patterns, especially when integrated with other deep learning architectures.

C. LSTM

LSTM is variant of RNN specifically intended to capture long-range dependencies in sequential manner, such as EEG signals. It uses memory cells and three gates (input, forget and output) for modulation of information flow through time. This scenario makes LSTM suitable for detecting temporal patterns that precede epileptic seizures.

The operation of an LSTM cell at time step is governed by the following equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (\text{Forget Gate})$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (\text{Input Gate})$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (\text{Candidate Cell State})$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (\text{Cell State Update})$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (\text{Output Gate})$$

$$h_t = o_t * \tanh(C_t) \quad (\text{Hidden Gate})$$

Here, x_t is input at time t , h_{t-1} is hidden state from previous step, and σ represents sigmoid activation function.

LSTM network has been employed to process sequential EEG features. Its ability to remember information over extended time intervals is critical for identifying subtle changes in brain activity that lead up to a seizure. LSTM networks contribute significantly to improving prediction accuracy in time-series biomedical data.

D. SqueezeNet

SqueezeNet is a compact CNN focused on high accuracy with a reduced number of parameters, which makes it very appropriate for real-time healthcare/embedded applications. Main idea is to use smaller convolutional filters (e.g. 1×1) in place of larger (e.g. 3×3) and use both squeeze and expand layers, collectively called fire modules.

A fire module includes:

- **squeeze layer** that incorporates 1×1 convolutions
- **expand layer** that employs a combination of 1×1 and 3×3 convolutions

The fire module is mathematically represented as:

$$\text{Squeeze}(x) = \text{ReLU}(W_s * x + b_s)$$

$$\text{Expand}(x) = \text{Concat}(\text{ReLU}(W_{e1} * x + b_{e1}), \text{ReLU}(W_{e3} * x + b_{e3}))$$

Where $*$ denotes convolution operation, and W , b are weights and biases for 1×1 and 3×3 convolutions.

SqueezeNet architecture is used for feature extraction from EEG signal representations. Due to its compact design and reduced memory footprint, it enables deployment on edge devices without compromising accuracy.

IV. RESULT DISCUSSION

A. DATASET

The CHB-MIT dataset was created through joint effort between Boston Children's Hospital and Massachusetts Institute of Technology (MIT) and made available on the PhysioNet platform. It includes long-term EEG recordings from 23 pediatric patients diagnosed with various types of epilepsy, including generalized epilepsy, temporal lobe epilepsy, and frontal lobe epilepsy. The patients' ages range from 1 month to 19 years, representing a diverse pediatric population with different seizure patterns and brain development stages [24].

Each patient's EEG was recorded continuously for approximately 23 hours, capturing both ictal (during seizure) and interictal (between seizures) intervals across sleep and wakefulness. The recordings include scalp EEG signals and, in some cases, subdural and stereo EEGs, obtained using different electrode configurations and tools. Data were sampled at variable frequencies ranging from 256 Hz to 400 Hz, depending on the specific recording setup. These recordings are stored in the European Data Format (EDF) and were later converted into MATLAB-compatible .mat files using the edfread function for efficient data handling and analysis.

A total of 23 electrodes were typically used for scalp EEG placement based on the international 10–20 system. The dataset structure facilitates segment-wise analysis, as the recordings are divided into 1-hour files for each subject. This

segmentation is particularly useful for machine learning-based seizure prediction models, which often require windowed data for training and evaluation.

The CHB-MIT dataset has been utilized extensively in previous studies for the development and benchmarking of seizure detection algorithms, providing a solid foundation for reproducible research in the field of neuroinformatics [25].

B. COMPUTATIONAL ANALYSIS

The output of this CNN layer is a tensor sized $256 \times 256 \times 32$, where the third dimension (32) is the number of feature maps generated by the layer. These feature maps contain key spatial features and are forwarded to the next layer of the CNN for deeper analysis and pattern learning.

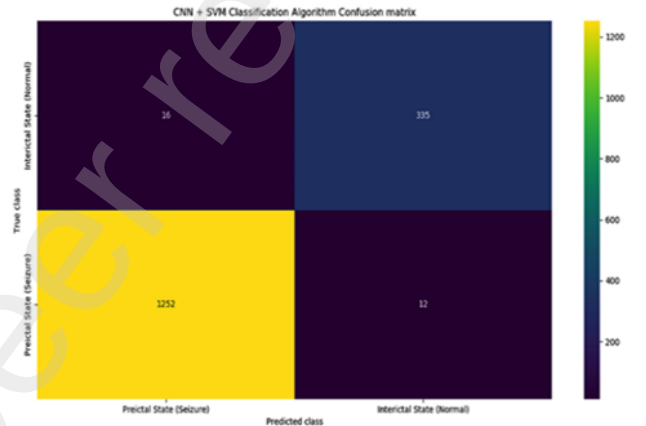


FIGURE 4. CNN-SVM Confusion Matrix.

Figure 4 shows a contrast between the proposed hybrid design and the current cutting-edge methods. Proposed hybrid was tested on basis of five metrics, namely: accuracy, precision, F1-score, sensitivity, and specificity. The hybrid model correctly classified 1,251 seizure cases and 341 non-seizure cases; it generated 13 false-positives and 10 false-negatives. This shows models classify EEG data very reliably. The confusion matrix analysis confirms its effectiveness in minimizing both missed detections and false alarms. Overall, the proposed approach outperforms existing techniques across all evaluated metrics.

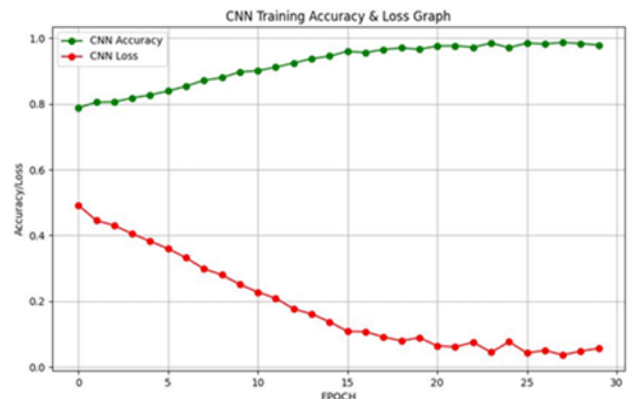


FIGURE 5. CNN training Accuracy and Loss Graph

Figure 5 shows how well the model performed with correct and incorrect predictions on the test samples that were unseen by the model. The model has excellent accuracy, and the model also performed with strength in real-time evaluation.

Notably, it predicted seizure events with an average lead time of 19 minutes. This is an improvement over existing models, which typically require 21 minutes. The shorter lead time allows earlier medical response, enhancing patient safety. These results affirm the model's practical utility in clinical seizure monitoring systems.

Model	Accur acy	Precis ion	Reca ll	F1- Score	Specifi city	Sensiti vity	MC C	NPV	FPR	FNR
CNN	0.9558 8	0.952 38	0.95 89	0.955 88	0.94 0.94	0.9577 5	0.94 118	0.95 161	0.06 341	0.07 954
LSTM	0.9302 3	0.921 05	0.94 34	0.921 05	0.94 0.94	0.9268 3	0.90 909	0.92 857	0.05 341	0.06 954
SqueezeNet	0.9464 3	0.933 33	0.94 118	0.94 0.94	0.9428 6	0.9166 7	0.94 34	0.95 192	0.07 741	0.08 454
ANN	0.9090 9	0.921 05	0.93 684	0.85 0.85	0.9417 5	0.8571 4	0.92 683	0.95 223	0.07 341	0.08 954

FIGURE 6. Comparison of performance metrics for training 70% and testing 30%

In a binary classification study, various machine learning models were evaluated using a data split where 70% was allocated for training and 30% for testing. CNN has the highest overall correctness of predictions, with an accuracy rating of 95.58%. When it comes to positive prediction accuracy, CNN likewise has the highest precision (95.28%) as per Figure 6.

Model	Accur acy	Precis ion	Reca ll	F1- Score	Specifi city	Sensiti vity	MC C	NPV	FPR	FNR
CNN	0.9615 4	0.954 55	0.96 296	0.958 9	0.9444 4	0.9594 6	0.95 2	0.95 238	0.06 001	0.07 023
LSTM	0.9508 2	0.948 28	0.94 34	0.921 05	0.9423 1	0.9268 3	0.90 909	0.92 857	0.05 341	0.06 954
SqueezeNet	0.9545 5	0.945 45	0.94 444	0.944 95	0.9428 6	0.9347 8	0.94 828	0.95 312	0.06 44	0.07 203
ANN	0.9302 3	0.942 31	0.94 783	0.863 64	0.9428 6	0.875 0.875	0.92 593	0.95 522	0.06 04	0.07 004

FIGURE 3. Comparison of performance metrics for training 80% and testing 20%

A similar comparison was conducted in Figure 7, but this time using an 80/20 train-test split. The same set of models. CNN has the greatest accuracy rating (96.15%), making their predictions the most accurate overall. CNN also has the highest precision (95.45%) in terms of positive prediction accuracy.

The comparative results clearly show that increasing the training data from 70% to 80% led to overall performance improvements across all models. CNN also benefited from more training data, showing an accuracy increase from 95.59% to 96.15% and precision from 95.24% to 95.45%. LSTM saw its accuracy jump from 93.02% to 95.08%,

while its precision remained stable. SqueezeNet improved in accuracy (94.64% to 95.45%) and precision (93.33% to 94.55%), reflecting better prediction strength. ANN showed significant growth, with accuracy rising from 90.91% to 93.02% and precision from 92.11% to 94.23%. In terms of F-Score, ANN rose from 85.00% to 86.36%. Sensitivity improved for ANN (87.50% to 94.78%) and remained steady for other models. Specificity and NPV also showed slight improvements for all models. Overall, training with more data led to stronger and more balanced model performance, especially for ANN.

V. CONCLUSION AND FUTURE WORK

We have implemented a deep learning-based technique for predicting epileptic seizures to improve detection performance and patient quality of life with our detection technique combining CNN-based feature extraction and machine learning classifiers perform better with respect to specificity and sensitivity than traditional methods. Inclusion of advanced preprocessing and a hybrid FOX-Chimp optimization strategy further strengthens the model's predictive capability. Comparative evaluations involving LSTM, ANN, CNN, and SqueezeNet models helped to identify unique advantages and limitations of each. While the proposed system performs effectively in patient-specific scenarios, there remains scope for optimizing preprocessing to improve signal quality and to reduce number of trainable parameters, in context of deep learning models. Future efforts should also focus on building generalized prediction models that can operate across diverse patient profiles without personalized training, moving closer to real-time, scalable solutions for clinical deployment.

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